

Minimal EFC Effective Field Theory Ansatz

Scalar-Tensor and Vector-Tensor Formulations
with Gravitational Slip from Entropy Gradients

Morten Magnusson
Energy-Flow Cosmology Initiative
ORCID: 0009-0002-4860-5095

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Abstract. We present two complementary effective field theory (EFT) ansätze for Energy-Flow Cosmology (EFC): a scalar-tensor formulation operationally equivalent to Horndeski/EFT-DE, and a vector-tensor formulation in which a dynamical 4-vector J^μ represents physical energy flow. Both are constructed so that the EFC sigmoid entropy field $S(a)$ governs the transition from a General Relativity (GR) regime at high redshift to a modified-gravity regime at intermediate redshift. We identify the three EFC target observables in terms of the gravitational slip convention used throughout, derive the schematic mechanism linking entropy-gradient stress to gravitational slip, and specify the consistency conditions required for stability and CMB compatibility. The scalar-tensor EFT is recommended as the immediate operational framework; the vector-tensor theory is identified as the ontologically natural EFC completion.

Introduction and Motivation

Energy-Flow Cosmology (EFC) posits that cosmological dynamics—including gravitational growth, structure formation, and cosmic acceleration—can be understood as emergent consequences of directed energy flow and entropy gradients across a dynamical grid. EFC treats cosmological evolution as a directed non-equilibrium flow rather than a purely passive background expansion, with mid-redshift physics ($0.5 \lesssim z \lesssim 1.5$) departing from General Relativity (GR) while the early universe ($z \gtrsim 10$) recovers standard cosmology automatically.

Phenomenological pipeline tests ($f\sigma_8$ fits, E_G predictions, and CMB-lensing constraints) motivate the following *target signature* that any EFC field theory must be capable of realising in some region of parameter space:

$$\mu(a) < 1, \tag{1}$$

$$\eta(a) > 1, \tag{2}$$

$$\Sigma(a) > 1. \tag{3}$$

These are defined and motivated in Section 2.

The purpose of this note is to construct the minimal EFT that can in principle produce this signature from EFC’s entropy-field structure, and to state carefully what has been derived versus what remains to be verified numerically.

Scope and status. This note does not represent canonical baseline EFC-D. It is a *proposed perturbation-sector EFT completion*, motivated by the Living Validation Ledger’s identified μ - Σ survival valley, the lensing barrier, and the documented $\text{GRAV} \rightarrow (\mu, \Sigma)$ structural gap that explicitly requires relativistic perturbation theory to close. The $\mu(k, z)$ and $\Sigma(k, z)$ functions treated here remain outside canonical baseline EFC-D pending that derivation.

Conventions and Observable Definitions

Metric perturbations

We work in the conformal Newtonian (longitudinal) gauge:

$$ds^2 = -(1 + 2\Psi) dt^2 + a^2(1 - 2\Phi) d\mathbf{x}^2. \quad (4)$$

In GR without anisotropic stress, $\Phi = \Psi$.

Gravitational slip

We adopt the standard convention

$$\eta \equiv \frac{\Phi}{\Psi}, \quad (5)$$

so that GR corresponds to $\eta = 1$. A value $\eta > 1$ means $|\Phi| > |\Psi|$: the lensing potential ($\Phi + \Psi$) is enhanced relative to the Newtonian potential Ψ . The quantity $\eta_{\text{slip}} \equiv \eta - 1$ is the slip *relative to GR*; throughout this note we use η (not η_{slip}) and state the GR value explicitly.

Modified-gravity functions

We use the standard set:

$$\mu(a, k) \equiv \frac{G_{\text{eff}}(a, k)}{G}, \quad (6)$$

$$\Sigma(a, k) \equiv \frac{G_{\text{lens}}(a, k)}{G} = \frac{\mu(a, k)}{2} (1 + \eta(a, k)). \quad (7)$$

Here μ controls the clustering of matter (entering the growth ODE), while Σ controls gravitational lensing. In GR: $\mu = \Sigma = \eta = 1$. Note that Eq. (7) uses $\eta \equiv \Phi/\Psi$ with GR value $\eta = 1$; this is the convention adopted throughout. The factor $(1 + \eta)/2 \rightarrow 1$ as $\eta \rightarrow 1$, recovering $\Sigma \rightarrow \mu$ in GR, and $\Sigma \rightarrow 1$ when also $\mu \rightarrow 1$.

EFC target signature

From pipeline analysis: $\delta_\mu \approx -0.075$ gives $\mu \approx 0.925$ in the transition regime. The Planck A_{lens} anomaly (observed lensing amplitude slightly exceeding the LCDM prediction, though not conclusively established) motivates exploring parameter regions with $\Sigma \gtrsim 1$. Combined with $\mu < 1$, this requires by Eq. (7):

$$\eta > \frac{2\Sigma}{\mu} - 1 > 1. \quad (8)$$

Concretely, if $\mu \approx 0.925$ and $\Sigma \approx 1.05$, then $\eta \approx 1.27$. This is the slip value the EFC field theory must generate in the transition regime while recovering $\eta \rightarrow 1$ at high z .

EFC Entropy Field

Both EFT formulations share the same background entropy field $S(a)$, defined as a sigmoid in log scale-factor:

$$S(a) = \frac{1}{1 + \exp\left[-\frac{\ln a - \ln a_t}{\Delta}\right]}, \quad (9)$$

where a_t is the transition scale factor and $\Delta > 0$ controls the sharpness of the transition. This field satisfies the EFC boundary conditions:

$$S(a) \rightarrow 0 \quad (a \rightarrow 0, \text{ early universe, GR limit}), \quad (10)$$

$$S(a) \rightarrow 1 \quad (a \rightarrow 1, \text{ today, maximum modification}), \quad (11)$$

$$\frac{dS}{d \ln a} \text{ peaks at } a = a_t. \quad (12)$$

From pipeline fits: $a_t \approx 0.30$ ($z_t \approx 2.3$), $\Delta \approx 0.3$.

The gravitational response function used in the growth ODE is:

$$\mu(a) = 1 + \delta_\mu S(a), \quad \delta_\mu \approx -0.075, \quad (13)$$

which gives $\mu \rightarrow 1$ as $S \rightarrow 0$ (high z), and $\mu < 1$ for $\delta_\mu < 0$ at low z . Note that Eq. (13) is a phenomenological ansatz for the growth ODE; the full $\mu(a, k)$ in the EFT will in general differ by scale-dependent corrections.

Ansatz I: Scalar-Tensor EFT

Action

We write the EFC scalar-tensor action in Horndeski/beyond-Horndeski form:

$$\mathcal{L}_{\text{EFC}}^{(S)} = \frac{M_{\text{Pl}}^2}{2} R + K(S, X) + G_3(S, X) \square S, \quad (14)$$

where $X \equiv -\frac{1}{2}(\nabla S)^2$ and K, G_3 are free functions to be fixed by EFC physics.

EFT Parameterisation (*ansatz*)

On a flat FLRW background, the action (14) maps to the Bellini–Sawicki α -parameter basis [1]:

$$\alpha_T(a) = 0 \quad (\text{imposed: required by GW170817}), \quad (15)$$

$$\alpha_B(a) \propto \frac{dS}{d \ln a} \quad (\text{ansatz: braiding from entropy gradient}), \quad (16)$$

$$\alpha_M(a) \propto S(a) \quad (\text{ansatz: running Planck mass from entropy field}). \quad (17)$$

The proportionality constants B_0 and M_0 are free parameters to be constrained by data; the functional forms in Eqs. (16)–(17) are motivated by EFC but are not derived from first principles here. The linear observable functions $\mu(a, k)$, $\eta(a, k)$, and $\Sigma(a, k)$ are then determined in the quasi-static limit by $\{\alpha_K, \alpha_B, \alpha_M, \alpha_T\}$ via expressions given in [1, 3]. We do *not* reproduce the full formulae here, as they are lengthy and the quasi-static expressions are subject to scale-dependent corrections; they must be evaluated numerically via `hi_class`.

EFC Identification and Expectations

With $\alpha_T = 0$ and the sigmoid driving α_B and α_M , the following qualitative behaviour is expected in some region of parameter space, but must be confirmed numerically:

- $\mu(a) < 1$ at mid- z , recovering $\mu \rightarrow 1$ at high z as $\alpha_B, \alpha_M \rightarrow 0$;
- non-trivial gravitational slip $\eta(a) > 1$ driven by $\alpha_B \neq 0$ (the sign follows from the quasi-static Bellini–Sawicki expressions for $B_0 > 0$; this is a qualitative expectation to be confirmed numerically);
- a region in $\{B_0, M_0\}$ space where $\Sigma > 1$ is simultaneously achievable.

Whether all three conditions hold together is a numerical question, not a consequence of the ansatz alone.

Ansatz II: Vector-Tensor EFT

Action

We write the EFC vector-tensor action with a dynamical 4-vector J^μ representing physical energy flow, coupled to the scalar entropy field:

$$\mathcal{L}_{\text{EFC}}^{(J)} = \frac{M_{\text{Pl}}^2}{2} R - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{m^2}{2} J_\mu J^\mu + \beta J^\mu \nabla_\mu S, \quad (18)$$

where $F_{\mu\nu} = \nabla_\mu J_\nu - \nabla_\nu J_\mu$ and β is the entropy-flow coupling constant. This action is motivated by the generalised Proca and Einstein-Aether classes [4].

Anisotropic Stress from Energy Flow

The energy-momentum tensor of the flow sector contains a gradient-energy contribution of the form:

$$T_{\mu\nu}^{(\nabla S)} = \beta^2 \left(\nabla_\mu S \nabla_\nu S - \frac{1}{4} g_{\mu\nu} (\nabla S)^2 \right). \quad (19)$$

In linear perturbation theory, the traceless spatial part of this tensor generates an anisotropic shear. Schematically, the scalar-mode shear stress from the perturbed entropy gradient takes the form:

$$\pi \sim \beta^2 \nabla \bar{S} \cdot \nabla \delta S, \quad (20)$$

where $\bar{S}(a)$ is the background sigmoid and δS is its perturbation. Via the perturbed Einstein equation $k^2(\Phi - \Psi) = 8\pi G a^2 \Pi$, one schematically expects a slip contribution of the form:

$$\eta - 1 \sim C_\eta \frac{\beta^2 (\nabla \bar{S})^2}{k^2 \rho_{\text{eff}}}, \quad C_\eta > 0 \text{ (to be determined)}, \quad (21)$$

where $C_\eta > 0$ is a dimensionless coefficient whose sign and magnitude are to be determined by the full linear perturbation analysis of the J^μ -sector; the positive sign is expected on physical grounds (entropy-gradient stress enhances the lensing potential relative to the Newtonian potential) but is not derived here. The estimate vanishes at high z where $\nabla \bar{S} \rightarrow 0$, recovering $\eta \rightarrow 1$ (GR) automatically.

Observable Summary

The vector-tensor ansatz connects the three EFC observables to the entropy-gradient mechanism through a single coupling β :

Observable	Source in ansatz	GR value	EFC target
$\mu(a)$	J^μ mass term modifies Poisson equation	1	$\mu \approx 0.925$ at $z \sim 0.7$
$\eta(a)$	$\pi \sim \beta^2(\nabla S)^2$ gives $\Phi \neq \Psi$	1	$\eta \gtrsim 1.2$ (schematic)
$\Sigma(a)$	$\Sigma = \mu(1 + \eta)/2$; net lensing from both effects	1	$\Sigma > 1$ (motivated by A_{lens})

Table 1: EFC observable signatures from the vector-tensor ansatz. All three arise from the entropy-gradient coupling β and vanish in the GR limit $S \rightarrow 0$ at high z . The target values for η and Σ are schematic pending a full perturbation analysis.

Comparison and Consistency Conditions

Property	Scalar-tensor EFT	Vector-tensor EFT
EFC ontology	Indirect (S as effective scalar)	Direct (J^μ as flow)
Free parameters	$\{B_0, M_0, a_t, \Delta\}$	$\{\beta, m^2, a_t, \Delta\}$
$\mu < 1$ mechanism	Via α_B, α_M	Via J^μ mass term
$\eta > 1$ mechanism	Via braiding $\alpha_B \neq 0$	Via $\pi \sim (\nabla S)^2$
$\Sigma > 1$	Numerical check required	Numerical check required
GR at high- z	Automatic ($\alpha \rightarrow 0$ as $S \rightarrow 0$)	Automatic ($\nabla \bar{S} \rightarrow 0$)
GW170817	$\alpha_T = 0$ imposed	Naturally small c_T correction
Numerical tools	hi_class, EFTCosmoMC	Requires new Boltzmann code
Immediate testability	High	Low

Table 2: Comparison of the two EFC-EFT ansatze.

Stability Conditions

For the scalar-tensor EFT, the standard no-ghost and no-gradient-instability conditions in the Horndeski class require:

$$Q_S \equiv \alpha_K + \frac{3}{2}\alpha_B^2 > 0 \quad (\text{no ghost}), \quad (22)$$

$$c_S^2 > 0 \quad (\text{no gradient instability}). \quad (23)$$

These must be verified numerically across the full sigmoid transition. For the vector-tensor EFT, the conditions $m^2 > 0$ and $\beta^2 < \beta_{\text{max}}^2$ (set by the gradient energy scale) are required to avoid tachyonic and ghost instabilities respectively.

CMB Compatibility

Both ansatze satisfy the primary CMB requirement $\mu \rightarrow 1, \eta \rightarrow 1$ at $z \gtrsim 10$ via the sigmoid suppression $S \rightarrow 0$. The Planck A_{lens} anomaly (observed lensing amplitude somewhat exceeding the Λ CDM prediction) motivates exploring parameter regions with $\Sigma \gtrsim 1$ in the range $z \sim 0.5$ – 1 . By Eq. (7), this requires:

$$\eta > \frac{2\Sigma}{\mu} - 1 \approx 1.24 \quad \text{for } \mu = 0.925, \Sigma = 1.05. \quad (24)$$

Whether the EFC ansatz achieve this is a numerical question pending full Boltzmann integration.

Discussion and Next Steps

The scalar-tensor EFT (Ansatz I) is the recommended immediate operational framework. It can be implemented in `hi_class` by specifying $\alpha_B(a) \propto dS/d\ln a$ and $\alpha_M(a) \propto S(a)$, running a joint CMB + BAO + RSD likelihood, and testing whether a parameter region exists satisfying $\mu < 1$, $\eta > 1$, $\Sigma > 1$ simultaneously consistent with Planck, BOSS, and DESI.

The vector-tensor EFT (Ansatz II) is the ontologically natural EFC completion, with physical energy flow J^μ as the fundamental degree of freedom. The connection between the two formulations is schematically:

$$S \sim \nabla_\mu J^\mu \quad \text{or} \quad S \sim |J|^2, \quad (25)$$

making the scalar a coarse-graining of the underlying flow field.

Falsification criterion. The scalar-tensor ansatz makes a testable structural prediction: a viable parameter region $\{B_0, M_0, a_t, \Delta\}$ must exist in which $\mu < 1$, $\eta > 1$, and $\Sigma > 1$ are simultaneously realised under the stability conditions $Q_S > 0$, $c_S^2 > 0$, and subject to CMB and large-scale structure constraints. If no such region exists in a full `hi_class` scan, the present scalar-tensor ansatz is *falsified as an EFC completion*, and the vector-tensor formulation or a qualitatively different EFT structure must be explored instead.

Open questions requiring numerical work:

1. Does a parameter region $\{B_0, M_0\}$ exist with $\mu < 1$, $\eta > 1$, $\Sigma > 1$ simultaneously in the scalar EFT?
2. Are the stability conditions $Q_S > 0$ and $c_S^2 > 0$ satisfied across the sigmoid transition?
3. What is the precise coefficient in the vector-tensor slip estimate Eq. (21)?
4. Does the J^μ action reduce to the scalar EFT in the isotropic, quasi-static limit?

Priority implementation steps:

1. Implement $\alpha_B(a)$, $\alpha_M(a)$ with EFC sigmoid parameterisation in `hi_class`.
2. Run joint Planck 2018 + DESI Y1 + BOSS likelihood to constrain $\{B_0, M_0, a_t, \Delta\}$.
3. Verify stability conditions numerically.
4. Derive linearised perturbation equations for J^μ -sector and compute $\eta(a, k)$ explicitly.

Summary

We have constructed two minimal EFT ansatzes for EFC, correcting and sharpening the conventions and claims of earlier drafts (v1–v5). The key points are:

- The gravitational slip is defined as $\eta \equiv \Phi/\Psi$ with GR value $\eta = 1$; the EFC target is $\eta > 1$ (not $\eta < 0$).

- The lensing function $\Sigma = \mu(1 + \eta)/2$ can exceed unity even with $\mu < 1$ when $\eta > 1$, consistent with the Planck A_{lens} motivation.
- The scalar-tensor EFT maps EFC to the Horndeski α -basis and is immediately testable via `hi_class`. The observable functions μ, η, Σ are driven by $\alpha_B \propto dS/d\ln a$ and $\alpha_M \propto S(a)$.
- The vector-tensor EFT provides the physically natural completion, with entropy-gradient stress $\pi \sim \beta^2(\nabla S)^2$ expected to generate slip $\eta > 1$ schematically. In scaling form: $\eta - 1 \sim C_\eta \beta^2(\nabla S)^2(k^2 \rho_{\text{eff}})^{-1}$ with $C_\eta > 0$ to be determined (Eq. 21). Full derivation of the perturbation equations remains as future work.
- Both formulations recover GR at high z automatically via $S \rightarrow 0$.

References

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